

TINGXI LI

800 W Campbell Road, Richardson, TX 75080

☎ (213)-795-5275 ✉ tingxi.li@utdallas.edu [in linkedin](#) [🏠 homepage](#) [🎓 Google Scholar](#)

Research Interests

Deep Learning Compilers, GPU Kernel Optimization, AI Systems, Compiler Robustness/Testing, AI Security

Education

University of Texas at Dallas

Aug. 2024 – Present

Doctor of Philosophy in Computer Science

Dalian University of Technology

Sep. 2019 – May 2024

Bachelor of Science

Publications

An Empirical Analysis of Compiler Bugs Across Tensor and Tile DSLs

Co-author

Under Review

- Analyzed and categorized compiler/code-generation bugs reported in multiple DSLs' GitHub issue trackers (including TileLang, Triton, and others). Led the taxonomy design for root causes, symptoms, and fixes, and performed a comparative analysis of bug patterns across tensor compilers, tile compilers, and traditional CPU compilers, highlighting both shared failure modes and compiler-specific differences.

The Paradigm Shift in AI Compilation in the Lens of Auto-Tuning

First Author

Under Review

- Systematized AI compiler auto-tuning through a multidimensional taxonomy to analyze the shift from schedule-centric to tile-centric compilation, concluding that while modern DSLs maximize usability, they still face significant scalability bottlenecks in supporting complex fused kernels and heterogeneous hardware compared to hand-tuned libraries.

SoK: Efficiency Robustness of Dynamic Deep Learning Systems

*Ravishka Rathnasuriya, **Tingxi Li**, Zexin Xu, Zihe Song, Mirazul Haque, Simin Chen, Wei Yang*

Usenix Sec'25

- Co-authored a systematization-of-knowledge (SoK) paper surveying efficiency attacks on dynamic DNN systems. Categorized attack strategies, reproduced published results, and benchmarked defenses in unstudied scenarios.

Identify then Exploit: Degrading Performance of Vision-based Deep Learning Systems

First Author

Under Review

- Proposed a targeted efficiency attack for vision-based dynamic deep learning (DynDL) pipeline systems. Our attack increases floating-point operations (FLOPs) and inference latency by up to 50× and 10× even under black-box setting.

Closed-loop Orchestration for Malicious Elicitation Techniques in Code Models

*Zexin Xu, **Tingxi Li**, et al.*

Technical Report

- Co-authored a public technical report detailing the attack and defense methodology, outlining surrogate model fine-tuning and iterative adversarial data integration pipeline.

Internship Experience

Sophgo

June. 2024 – Aug. 2024

Research Intern

Shenzhen, China

- Developed and implemented API code in C++ for models deployed on RISC-V processors. Authored comprehensive documentation to support efficient deployment and usage.
- Fine-tuned models on private datasets, identified failure cases, and implemented data augmentation strategies.

Projects

Amazon Trusted AI Competition

Nov. 2024 – Jul. 2025

Team Member

Seattle, WA

- Finalist team in the Amazon Trusted AI Competition, focused on jailbreaking CodeLLMs under a black-box setting, awarded \$250K prize.
- Fine-tuned surrogate models to evaluate and guide adversarial prompt generation, achieving 0.80+ recall and 0.75+ F1-score in malicious intent detection.

Adversarial Attack on a Robotic Arm

Feb. 2025 – Present

UT Dallas

Dallas, TX

- Analyzed robustness in Bulletarm grasping tasks, revealing deterministic environment dynamics and non-robust action-space boundaries that can be exploited by adversarial environment settings.